

Project Signal and Image Processing . Mod 1

Project members:

- Tito Nicola Drugman, 502252, titoncola.drugman01@universitadipavia.it
- Angelica Pagni, 505887, angelica.pagni01@universitadipavia.it

```
rng(48);

subfolders = {'underexposed', 'correctly_exposed', 'overexposed'};
output_folder = 'gamma_corrected';

imageData = [];
labels = [];

% loop through each subfolder to read images
for i = 1:length(subfolders)
    folderPath = subfolders{i};

    imgFiles = dir(fullfile(folderPath, '*.jpg'));

    for j = 1:length(imgFiles)
        imgPath = fullfile(folderPath, imgFiles(j).name);

        img = imread(imgPath);

        % convert image to YCbCr and extract the Y channel
        imgYCbCr = rgb2ycbcr(img);
        Y = imgYCbCr(:,:,1);

        % calculate and normalize histogram
        histY = imhist(Y);
        histY = histY / sum(histY);

        % extract features: mean, variance, and skewness
        meanY = mean(histY);
        varY = var(histY);
        skewY = skewness(double(histY));

        % count pixels in the min and max bins
        minYValue = min(Y(:)); % min intensity value in Y channel
        maxYValue = max(Y(:)); % max intensity value in Y channel
        minBinCount = sum(Y(:) == minYValue);
        maxBinCount = sum(Y(:) == maxYValue);

        imageData = [imageData; meanY, varY, skewY, minBinCount, maxBinCount];
        labels = [labels; i];
    end
end
```

```

        % encoding: 1-underexposed, 2-correctly_exposed, 3-overexposed
    end
end

```

Train-test split

```

splitRatio = 0.8; % 80% for training, 20% for testing
uniqueLabels = unique(labels);
numLabels = numel(uniqueLabels);
XTrain = [];
XTest = [];
YTrain = [];
YTest = [];

for i = 1:numLabels
    idx = find(labels == uniqueLabels(i));
    numDataPoints = numel(idx);
    numTrain = floor(splitRatio * numDataPoints);
    shuffledIndices = idx(randperm(numDataPoints));
    trainIdx = shuffledIndices(1:numTrain);
    testIdx = shuffledIndices(numTrain+1:end);
    XTrain = [XTrain; imageData(trainIdx,:)];
    YTrain = [YTrain; labels(trainIdx)];
    XTest = [XTest; imageData(testIdx,:)];
    YTest = [YTest; labels(testIdx)];
end

```

Normalize

```

%normalize the features
%min_vals = min(XTrain(:, 1:5), [], 1);
%max_vals = max(XTrain(:, 1:5), [], 1);
%XTrain(:, 1:5) = (XTrain(:, 1:5) - min_vals) ./ (max_vals - min_vals);
%XTest(:, 1:5) = (XTest(:, 1:5) - min_vals) ./ (max_vals - min_vals);

```

KNN

```

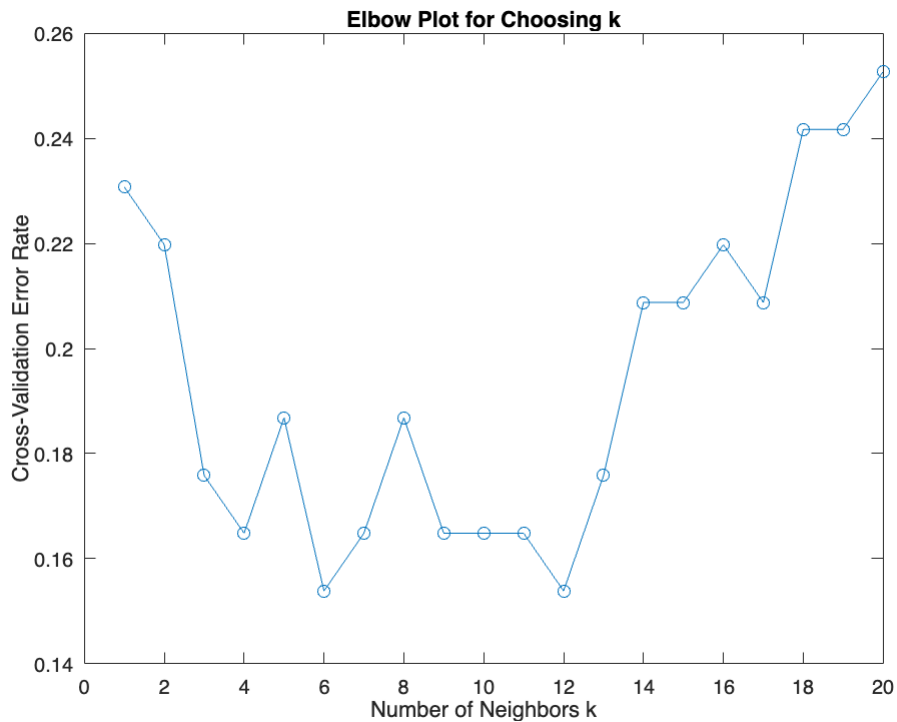
% KNN plot to find the optimal number of neighbors
kRange = 1:20;
errorRates = zeros(length(kRange), 1);

for k = kRange
    model = fitcknn(XTrain, YTrain, 'NumNeighbors', k, 'CrossVal', 'on');
    errorRates(k) = kfoldLoss(model);
end

figure;
plot(kRange, errorRates, '-o');
xlabel('Number of Neighbors k');
ylabel('Cross-Validation Error Rate');

```

```
title('Elbow Plot for Choosing k');
```



```
% optimal k
[~, optimalK] = min(errorRates);
disp(['Optimal number of neighbors: ', num2str(optimalK)]);
```

Optimal number of neighbors: 6

```
% -----

c = cvpartition(YTrain, 'KFold', 10);

% train KNN model with cross-validation using optimal NN
knnCVModel = fitcknn(XTrain, YTrain, 'CVPartition', c, 'NumNeighbors', optimalK);

% cross-validation loss for each fold
foldLosses = kfoldLoss(knnCVModel, 'Mode', 'individual');

% find fold with minimum loss
[~, bestFold] = min(foldLosses);
knncross = knnCVModel.Trained{bestFold};

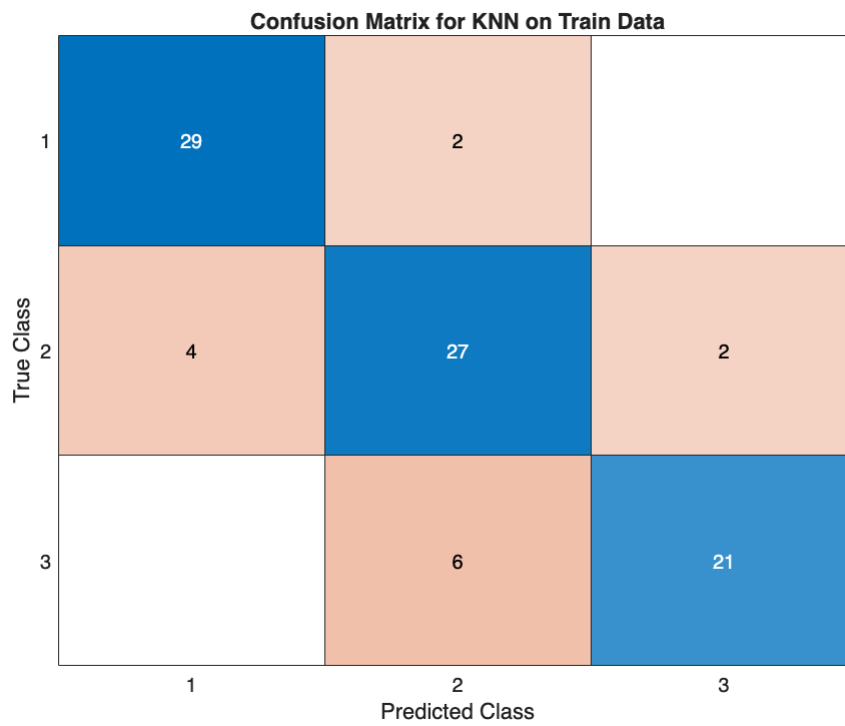
% -----

% evaluate on training data and test data
predictedLabels = predict(knncross, XTrain);
accuracy = sum(predictedLabels == YTrain) / numel(YTrain);
```

```
disp(['Train Accuracy KNN: ', num2str(accuracy * 100), '%']);
```

Train Accuracy KNN: 84.6154%

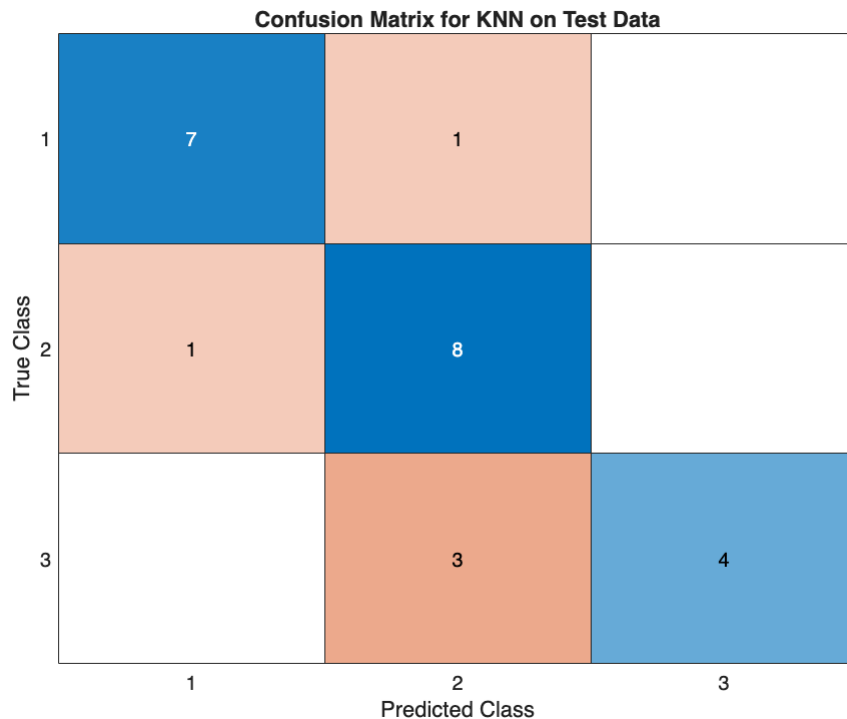
```
confusionchart(YTrain, predictedLabels, 'Title', 'Confusion Matrix for KNN on Train Data');
```



```
predictedLabels_test_knn = predict(knncross, XTest);  
accuracy_test_knn = sum(predictedLabels_test_knn == YTest) / numel(YTest);  
disp(['Test Accuracy KNN: ', num2str(accuracy_test_knn * 100), '%']);
```

Test Accuracy KNN: 79.1667%

```
figure;  
confusionchart(YTest, predictedLabels_test_knn, 'Title', 'Confusion Matrix for KNN on Test Data');
```



Decision Tree

```
c_tree = cvpartition(YTrain, 'KFold', 10);

% train Decision Tree model
treeCVModel = fitctree(XTrain, YTrain, 'CVPartition', c_tree);

% calculate cross-validation loss for each fold
foldLosses = kfoldLoss(treeCVModel, 'Mode', 'individual');

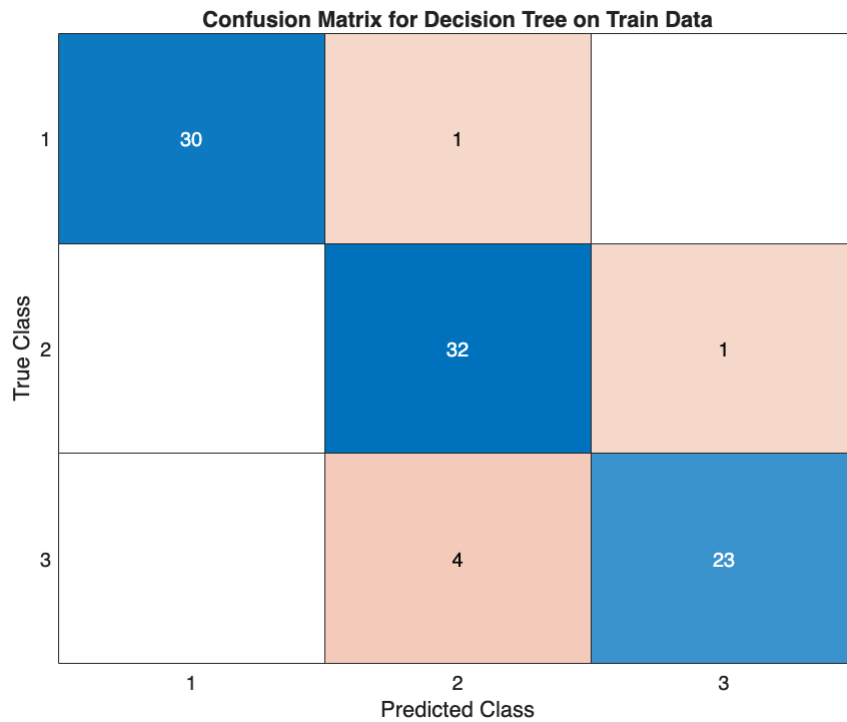
% find the fold with the minimum loss (best performance)
[~, bestFold] = min(foldLosses);
bestTreeModel = treeCVModel.Trained{bestFold};

% -----

% evaluate performance on training data and test data
predictedLabels_train_tree = predict(bestTreeModel, XTrain);
accuracy_tree = sum(predictedLabels_train_tree == YTrain) / numel(YTrain);
disp(['Train Accuracy tree: ', num2str(accuracy_tree * 100), '%']);
```

Train Accuracy tree: 93.4066%

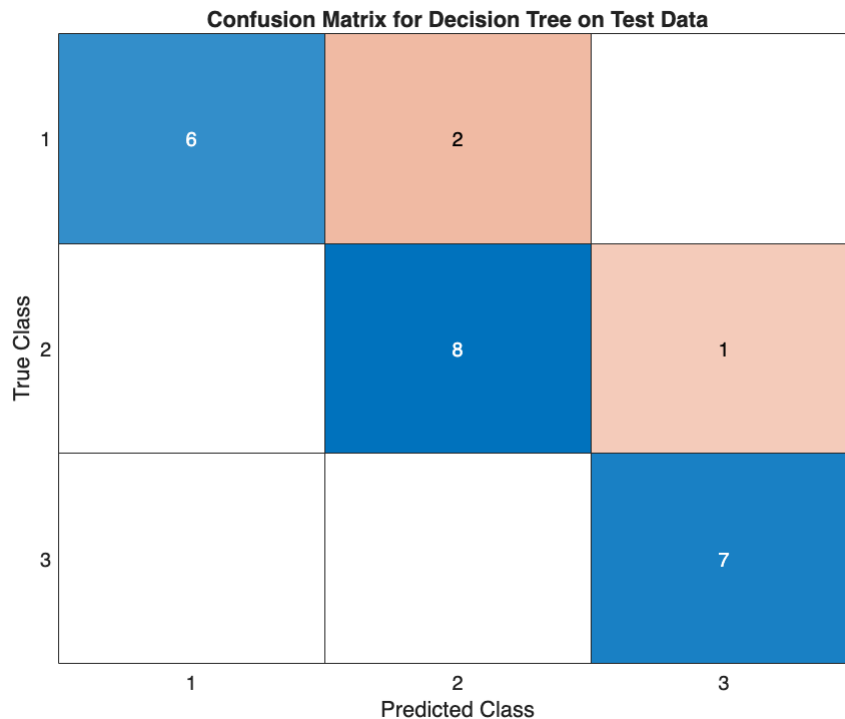
```
confusionchart(YTrain, predictedLabels_train_tree, 'Title', 'Confusion Matrix for  
Decision Tree on Train Data');
```



```
predictedLabels_test_tree = predict(bestTreeModel, XTest);  
accuracy_test_tree = sum(predictedLabels_test_tree == YTest) / numel(YTest);  
disp(['Test Accuracy Decision Tree: ', num2str(accuracy_test_tree * 100), '%']);
```

Test Accuracy Decision Tree: 87.5%

```
figure;  
confusionchart(YTest, predictedLabels_test_tree, 'Title', 'Confusion Matrix for  
Decision Tree on Test Data');
```



Compute metrics for correctly exposed images

```
correctlyExposedFolderPath = fullfile(subfolders{2}, '*.jpg');

imgFiles = dir(correctlyExposedFolderPath);

numFiles = numel(imgFiles);
norm_means = zeros(numFiles, 1);
norm_vars = zeros(numFiles, 1);
norm_skew = zeros(numFiles, 1);
norm_minBins = zeros(numFiles, 1);
norm_maxBins = zeros(numFiles, 1);

for j = 1:numFiles
    imgPath = fullfile(imgFiles(j).folder, imgFiles(j).name);
    img = imread(imgPath);
    imgYCbCr = rgb2ycbcr(im2double(img));
    Y = imgYCbCr(:,:,1);

    % compute metrics
    norm_means(j) = mean(Y, 'all');
    norm_vars(j) = var(Y, 0, 'all');
    norm_skew(j) = skewness(double(Y(:)));
    minYValue = min(Y(:));
    maxYValue = max(Y(:));
    norm_minBins(j) = sum(Y(:) == minYValue);
    norm_maxBins(j) = sum(Y(:) == maxYValue);
end
```

```

target_mean = mean(norm_means);
target_variance = mean(norm_vars);
target_skew = mean(norm_skew);
target_minBinCount = mean(norm_minBins);
target_maxBinCount = mean(norm_maxBins);

% scale
norm_target_minBinCount = target_minBinCount / max(target_minBinCount,
target_maxBinCount);
norm_target_maxBinCount = target_maxBinCount / max(target_minBinCount,
target_maxBinCount);

disp(['Target Mean: ', num2str(target_mean)]);

```

Target Mean: 0.50782

```
disp(['Target Variance: ', num2str(target_variance)]);
```

Target Variance: 0.02893

```
disp(['Target Skewness: ', num2str(target_skew)]);
```

Target Skewness: -0.22396

```
disp(['Target Min Bin Count: ', num2str(norm_target_minBinCount)]);
```

Target Min Bin Count: 0.030701

```
disp(['Target Max Bin Count: ', num2str(norm_target_maxBinCount)]);
```

Target Max Bin Count: 1

```

target_minBinCount = norm_target_minBinCount;
target_maxBinCount = norm_target_maxBinCount;

```

```
disp('-----');
```

```
disp('---- GAMMA CORRECTION ----');
```

---- GAMMA CORRECTION ----

```
disp('-----');
```

```
function img_corrected = applyGammaCorrection(img, gamma)
```



```

    img_corrected = img .^ gamma;
end

function error = compute_gamma_error(gamma, img, target_mean, target_variance,
target_skew, target_minBinCount, target_maxBinCount)
    img_corrected = applyGammaCorrection(img, gamma);
    mean_corrected = mean(img_corrected, 'all');
    var_corrected = var(img_corrected, 0, 'all');
    skew_corrected = skewness(double(img_corrected(:)));
    minYValue = min(img_corrected(:));
    maxYValue = max(img_corrected(:));
    minBinCount_corrected = sum(img_corrected(:) == minYValue);
    maxBinCount_corrected = sum(img_corrected(:) == maxYValue);
    error = (mean_corrected - target_mean)^2 + (var_corrected - target_variance)^2
+ ...
        0.01*(skew_corrected - target_skew)^2 + 0.01*(minBinCount_corrected -
target_minBinCount)^2 + ...
        0.01*(maxBinCount_corrected - target_maxBinCount)^2;
end

function best_gamma = find_best_gamma(img, target_mean, target_variance,
target_skew, target_minBinCount, target_maxBinCount, lower_bound, upper_bound)
    gamma_init = 1;
    options = optimoptions('fmincon', 'Display', 'off');
    best_gamma = fmincon(@(gamma) compute_gamma_error(gamma, img, target_mean,
target_variance, target_skew, target_minBinCount, target_maxBinCount), gamma_init,
[], [], [], [], lower_bound, upper_bound, [], options);
end

if ~exist(output_folder, 'dir')
    mkdir(output_folder);
end

for k = 1:length(subfolders)
    if ~exist(fullfile(output_folder, subfolders{k}), 'dir')
        mkdir(fullfile(output_folder, subfolders{k}));
    end
end

for i = 1:length(subfolders)
    folderPath = subfolders{i};
    imgFiles = dir(fullfile(folderPath, '*.jpg'));
    for j = 1:length(imgFiles)
        imgPath = fullfile(folderPath, imgFiles(j).name);
        img = im2double(imread(imgPath));
        img_ycbcr = rgb2ycbcr(img);
    end
end

```

```

img_y = img_ycbcr(:,:,1);
Cb = img_ycbcr(:,:,2);
Cr = img_ycbcr(:,:,3);

% gamma bounds based on image type
if i == 1 % underexposed
    gamma_lb = 0.1;
    gamma_ub = 1;
elseif i == 2 % correctly_exposed
    gamma_lb = 0.8;
    gamma_ub = 1.2;
else % overexposed
    gamma_lb = 1.1;
    gamma_ub = 5;
end

best_gamma = find_best_gamma(img_y, target_mean, target_variance,
target_skew, target_minBinCount, target_maxBinCount, gamma_lb, gamma_ub);
y_corrected = applyGammaCorrection(img_y, best_gamma);
img_recombined = cat(3, y_corrected, Cb, Cr);
img_correct = ycbcr2rgb(img_recombined);
[~, name, ext] = fileparts(imgPath);
output_file = fullfile(output_folder, subfolders{i}, [name, '_correct',
ext]);
imwrite(im2uint8(img_correct), output_file);
end
end

```

Evaluate

```

labels_correct = [];
filepaths_correct = [];
for i = 1:length(subfolders)
    files_correct = dir(fullfile(output_folder, subfolders{i}, '*.jpg'));
    filepaths_correct = [filepaths_correct; cellfun(@(s) fullfile(output_folder,
subfolders{i}, s), {files_correct.name}', 'UniformOutput', false)];
    labels_correct = [labels_correct; repmat(i, length(files_correct), 1)];
end

features_correct = [];
for i = 1:numel(filepaths_correct)
    img = imread(filepaths_correct{i});
    imgYCbCr = rgb2ycbcr(im2double(img));
    Y = imgYCbCr(:,:,1);
    histY = imhist(Y);
    histY = histY / sum(histY);
    meanY = mean(histY);
    varY = var(histY);
    skewY = skewness(double(histY));

```

```

minYValue = min(Y(:));
maxYValue = max(Y(:));
minBinCount = sum(Y(:) == minYValue);
maxBinCount = sum(Y(:) == maxYValue);
features_correct = [features_correct; meanY, varY, skewY, minBinCount,
maxBinCount];
end

```

Final Results

```

knnCounts = zeros(3, 1);
treeCounts = zeros(3, 1);

disp('Starting image classification...');

```

Starting image classification...

```

for i = 1:length(subfolders)
    folderPath = fullfile(output_folder, subfolders{i});
    imgFiles = dir(fullfile(folderPath, '*.jpg'));

    disp(['Processing ', subfolders{i}, '...']);

    for j = 1:length(imgFiles)
        imgPath = fullfile(folderPath, imgFiles(j).name);
        img = imread(imgPath);
        imgYCbCr = rgb2ycbcr(img);
        Y = imgYCbCr(:,:,1);

        histY = imhist(Y) / sum(imhist(Y));
        meanY = mean(histY);
        varY = var(histY);
        skewY = skewness(double(histY));
        minYValue = min(Y(:));
        maxYValue = max(Y(:));
        minBinCount = sum(Y(:) == minYValue);
        maxBinCount = sum(Y(:) == maxYValue);
        features = [meanY, varY, skewY, minBinCount, maxBinCount];

        % KNN classification
        knnLabel = predict(knncross, features);
        knnCounts(knnLabel) = knnCounts(knnLabel) + 1;
        fprintf('KNN: %s classified as %s\n', imgFiles(j).name,
subfolders{knnLabel});

        % Decision Tree classification
        treeLabel = predict(bestTreeModel, features);
        treeCounts(treeLabel) = treeCounts(treeLabel) + 1;
        fprintf('Decision Tree: %s classified as %s\n', imgFiles(j).name,
subfolders{treeLabel});
    end
end

```

```
end
end
```

Processing underexposed...

[illegible]

KNN: IMG-20240702-WA0062_correct.jpg classified as underexposed
 Decision Tree: IMG-20240702-WA0062_correct.jpg classified as underexposed
 KNN: IMG-20240702-WA0064_correct.jpg classified as correctly_exposed
 Decision Tree: IMG-20240702-WA0064_correct.jpg classified as correctly_exposed
 KNN: img10_correct.jpg classified as underexposed
 Decision Tree: img10_correct.jpg classified as correctly_exposed
 KNN: img13_correct.jpg classified as underexposed
 Decision Tree: img13_correct.jpg classified as underexposed
 KNN: img15_correct.jpg classified as underexposed
 Decision Tree: img15_correct.jpg classified as underexposed
 KNN: img16_correct.jpg classified as underexposed
 Decision Tree: img16_correct.jpg classified as correctly_exposed
 KNN: img1_correct.jpg classified as underexposed
 Decision Tree: img1_correct.jpg classified as underexposed
 KNN: img24_correct.jpg classified as underexposed
 Decision Tree: img24_correct.jpg classified as underexposed
 KNN: img25_correct.jpg classified as underexposed
 Decision Tree: img25_correct.jpg classified as underexposed
 KNN: img26_correct.jpg classified as underexposed
 Decision Tree: img26_correct.jpg classified as underexposed
 KNN: img28_correct.jpg classified as underexposed
 Decision Tree: img28_correct.jpg classified as correctly_exposed
 KNN: img29_correct.jpg classified as underexposed
 Decision Tree: img29_correct.jpg classified as underexposed
 KNN: img32_correct.jpg classified as underexposed
 Decision Tree: img32_correct.jpg classified as underexposed
 KNN: img33_correct.jpg classified as correctly_exposed
 Decision Tree: img33_correct.jpg classified as correctly_exposed
 KNN: img34_correct.jpg classified as correctly_exposed
 Decision Tree: img34_correct.jpg classified as correctly_exposed
 KNN: img35_correct.jpg classified as correctly_exposed
 Decision Tree: img35_correct.jpg classified as correctly_exposed
 KNN: img36_correct.jpg classified as correctly_exposed
 Decision Tree: img36_correct.jpg classified as correctly_exposed
 KNN: img37_correct.jpg classified as correctly_exposed
 Decision Tree: img37_correct.jpg classified as correctly_exposed
 KNN: img38_correct.jpg classified as correctly_exposed
 Decision Tree: img38_correct.jpg classified as correctly_exposed
 KNN: img39_correct.jpg classified as overexposed
 Decision Tree: img39_correct.jpg classified as correctly_exposed
 KNN: img3_correct.jpg classified as underexposed
 Decision Tree: img3_correct.jpg classified as underexposed
 KNN: img9_correct.jpg classified as correctly_exposed
 Decision Tree: img9_correct.jpg classified as correctly_exposed

```
disp('Classification complete.');
```

Classification complete.

```
disp('KNN Classification Results:');
```

KNN Classification Results:

```
disp(knnCounts);
```

```

33
75
7

```

```
disp('Decision Tree Classification Results:');
```

Decision Tree Classification Results:

```
disp(treeCounts);
```

```
17
89
9
```

```
disp('KNN Classification Summary:');
```

KNN Classification Summary:

```
fprintf('Number of images classified as correctly exposed: %d\n', knnCounts(2));
```

Number of images classified as correctly exposed: 75

```
fprintf('Number of images classified as overexposed: %d\n', knnCounts(3));
```

Number of images classified as overexposed: 7

```
fprintf('Number of images classified as underexposed: %d\n', knnCounts(1));
```

Number of images classified as underexposed: 33

```
disp('Decision Tree Classification Summary:');
```

Decision Tree Classification Summary:

```
fprintf('Number of images classified as correctly exposed: %d\n', treeCounts(2));
```

Number of images classified as correctly exposed: 89

```
fprintf('Number of images classified as overexposed: %d\n', treeCounts(3));
```

Number of images classified as overexposed: 9

```
fprintf('Number of images classified as underexposed: %d\n', treeCounts(1));
```

Number of images classified as underexposed: 17

```
disp('-----');
```

```
disp('----- BONUS POINT -----');
```

----- BONUS POINT -----

```
disp('----- 1st approach -----');
```

----- 1st approach -----


```
disp('-----');
```

```
-----
```

```
niqe_scores_original = [];  
niqe_scores_corrected = [];  
improvement_count = 0;  
  
for i = 1:length(subfolders)  
    originalFolderPath = subfolders{i};  
    correctedFolderPath = fullfile(output_folder, subfolders{i});  
  
    originalImgFiles = dir(fullfile(originalFolderPath, '*.jpg'));  
    correctedImgFiles = dir(fullfile(correctedFolderPath, '*.jpg'));  
  
    for j = 1:length(originalImgFiles)  
        originalImgPath = fullfile(originalFolderPath, originalImgFiles(j).name);  
        correctedImgName = strrep(originalImgFiles(j).name, '.jpg', '_correct.jpg');  
        correctedImgPath = fullfile(correctedFolderPath, correctedImgName);  
  
        if exist(correctedImgPath, 'file')  
            img_original = im2double(imread(originalImgPath));  
            img_corrected = im2double(imread(correctedImgPath));  
  
            original_niqe = niqe(img_original);  
            corrected_niqe = niqe(img_corrected);  
  
            niqe_scores_original = [niqe_scores_original; original_niqe];  
            niqe_scores_corrected = [niqe_scores_corrected; corrected_niqe];  
  
            if corrected_niqe < original_niqe  
                improvement_count = improvement_count + 1;  
            end  
  
            fprintf('%s - Original NIQE: %f, Corrected NIQE: %f\n',  
originalImgFiles(j).name, original_niqe, corrected_niqe);  
        else  
            fprintf('Corrected image not found: %s\n', correctedImgPath);  
        end  
    end  
end  
end
```

```
IMG-20240702-WA0022.jpg - Original NIQE: 3.880886, Corrected NIQE: 3.905780  
IMG-20240702-WA0024.jpg - Original NIQE: 4.421346, Corrected NIQE: 4.440012  
IMG-20240702-WA0028.jpg - Original NIQE: 4.921907, Corrected NIQE: 4.879202  
IMG-20240702-WA0031.jpg - Original NIQE: 3.904665, Corrected NIQE: 4.506142  
IMG-20240702-WA0034.jpg - Original NIQE: 5.112049, Corrected NIQE: 5.231965  
IMG-20240702-WA0037.jpg - Original NIQE: 4.992699, Corrected NIQE: 5.184564  
IMG-20240702-WA0040.jpg - Original NIQE: 4.846499, Corrected NIQE: 4.908783  
IMG-20240702-WA0043.jpg - Original NIQE: 4.810751, Corrected NIQE: 4.754339
```

IMG-20240702-WA0046.jpg - Original NIQE: 4.481009, Corrected NIQE: 4.582852
IMG-20240702-WA0049.jpg - Original NIQE: 4.016109, Corrected NIQE: 4.143540
IMG-20240702-WA0051.jpg - Original NIQE: 4.418931, Corrected NIQE: 4.564013
IMG-20240702-WA0054.jpg - Original NIQE: 4.469673, Corrected NIQE: 4.484898
IMG-20240702-WA0057.jpg - Original NIQE: 4.299667, Corrected NIQE: 4.515714
IMG-20240702-WA0058.jpg - Original NIQE: 4.935446, Corrected NIQE: 5.076668
IMG-20240702-WA0063.jpg - Original NIQE: 7.019991, Corrected NIQE: 6.913203
img1.jpg - Original NIQE: 3.641652, Corrected NIQE: 3.698330
img11.jpg - Original NIQE: 4.137483, Corrected NIQE: 4.434336
img12.jpg - Original NIQE: 4.668748, Corrected NIQE: 4.705270
img13.jpg - Original NIQE: 2.920321, Corrected NIQE: 2.926066
img14.jpg - Original NIQE: 3.791849, Corrected NIQE: 4.017700
img15.jpg - Original NIQE: 4.339508, Corrected NIQE: 4.392828
img19.jpg - Original NIQE: 3.858264, Corrected NIQE: 3.863437
img2.jpg - Original NIQE: 6.009334, Corrected NIQE: 5.945499
img20.jpg - Original NIQE: 5.033616, Corrected NIQE: 5.072199
img25.jpg - Original NIQE: 3.282256, Corrected NIQE: 3.574527
img26.jpg - Original NIQE: 2.865517, Corrected NIQE: 2.973515
img27.jpg - Original NIQE: 4.981803, Corrected NIQE: 5.071695
img28.jpg - Original NIQE: 4.818335, Corrected NIQE: 4.853440
img29.jpg - Original NIQE: 2.690832, Corrected NIQE: 2.748640
img30.jpg - Original NIQE: 4.475939, Corrected NIQE: 4.504781
img31.jpg - Original NIQE: 5.262543, Corrected NIQE: 5.167870
img32.jpg - Original NIQE: 3.740043, Corrected NIQE: 3.611842
img33.jpg - Original NIQE: 4.426218, Corrected NIQE: 4.423479
img34.jpg - Original NIQE: 6.323544, Corrected NIQE: 6.202548
img35.jpg - Original NIQE: 4.534404, Corrected NIQE: 4.565066
img37.jpg - Original NIQE: 4.267354, Corrected NIQE: 4.299858
img38.jpg - Original NIQE: 4.228931, Corrected NIQE: 4.878118
img39.jpg - Original NIQE: 3.095441, Corrected NIQE: 3.102619
img8.jpg - Original NIQE: 3.935611, Corrected NIQE: 4.023382
IMG-20240702-WA0023.jpg - Original NIQE: 3.478348, Corrected NIQE: 3.476137
IMG-20240702-WA0026.jpg - Original NIQE: 3.305621, Corrected NIQE: 3.285692
IMG-20240702-WA0029.jpg - Original NIQE: 4.215973, Corrected NIQE: 4.217099
IMG-20240702-WA0032.jpg - Original NIQE: 3.652855, Corrected NIQE: 3.704913
IMG-20240702-WA0035.jpg - Original NIQE: 3.960314, Corrected NIQE: 3.948036
IMG-20240702-WA0038.jpg - Original NIQE: 4.152542, Corrected NIQE: 4.130998
IMG-20240702-WA0041.jpg - Original NIQE: 3.997599, Corrected NIQE: 3.900842
IMG-20240702-WA0044.jpg - Original NIQE: 3.710942, Corrected NIQE: 3.696272
IMG-20240702-WA0047.jpg - Original NIQE: 4.511117, Corrected NIQE: 4.536040
IMG-20240702-WA0050.jpg - Original NIQE: 3.362707, Corrected NIQE: 3.357875
IMG-20240702-WA0053.jpg - Original NIQE: 3.797067, Corrected NIQE: 3.845991
IMG-20240702-WA0056.jpg - Original NIQE: 3.546522, Corrected NIQE: 3.545472
IMG-20240702-WA0060.jpg - Original NIQE: 4.395245, Corrected NIQE: 4.403255
IMG-20240702-WA0061.jpg - Original NIQE: 3.468843, Corrected NIQE: 3.509886
IMG-20240702-WA0065.jpg - Original NIQE: 3.974133, Corrected NIQE: 3.936793
img10.jpg - Original NIQE: 1.972172, Corrected NIQE: 1.971973
img12.jpg - Original NIQE: 3.628249, Corrected NIQE: 3.602581
img13.jpg - Original NIQE: 2.677833, Corrected NIQE: 2.670335
img14.jpg - Original NIQE: 3.875328, Corrected NIQE: 3.875646
img15.jpg - Original NIQE: 3.890002, Corrected NIQE: 3.889520
img18.jpg - Original NIQE: 4.023543, Corrected NIQE: 3.970956
img19.jpg - Original NIQE: 3.028174, Corrected NIQE: 3.039792
img21.jpg - Original NIQE: 3.312819, Corrected NIQE: 3.311882
img24.jpg - Original NIQE: 4.117466, Corrected NIQE: 4.118570
img25.jpg - Original NIQE: 3.261810, Corrected NIQE: 3.254122
img27.jpg - Original NIQE: 4.399133, Corrected NIQE: 4.277757
img29.jpg - Original NIQE: 2.016839, Corrected NIQE: 2.012642
img30.jpg - Original NIQE: 3.842576, Corrected NIQE: 3.857621
img31.jpg - Original NIQE: 4.192754, Corrected NIQE: 4.298316
img32.jpg - Original NIQE: 4.266108, Corrected NIQE: 4.173285
img33.jpg - Original NIQE: 3.523412, Corrected NIQE: 3.648096
img34.jpg - Original NIQE: 3.111594, Corrected NIQE: 4.144613
img35.jpg - Original NIQE: 3.900316, Corrected NIQE: 4.143225

```

img36.jpg - Original NIQE: 3.090441, Corrected NIQE: 2.932768
img37.jpg - Original NIQE: 3.908488, Corrected NIQE: 4.025197
img38.jpg - Original NIQE: 4.181540, Corrected NIQE: 4.434678
img39.jpg - Original NIQE: 3.114943, Corrected NIQE: 2.933983
img4.jpg - Original NIQE: 3.603658, Corrected NIQE: 3.608367
img5.jpg - Original NIQE: 4.328086, Corrected NIQE: 4.337178
img6.jpg - Original NIQE: 4.509018, Corrected NIQE: 4.518391
img7.jpg - Original NIQE: 3.654011, Corrected NIQE: 3.656952
img8.jpg - Original NIQE: 4.021414, Corrected NIQE: 4.035658
IMG-20240702-WA0025.jpg - Original NIQE: 3.550572, Corrected NIQE: 3.744254
IMG-20240702-WA0027.jpg - Original NIQE: 6.119376, Corrected NIQE: 5.658124
IMG-20240702-WA0030.jpg - Original NIQE: 4.436063, Corrected NIQE: 4.376727
IMG-20240702-WA0033.jpg - Original NIQE: 3.960667, Corrected NIQE: 4.066839
IMG-20240702-WA0036.jpg - Original NIQE: 3.557605, Corrected NIQE: 3.696085
IMG-20240702-WA0039.jpg - Original NIQE: 4.213604, Corrected NIQE: 4.100835
IMG-20240702-WA0042.jpg - Original NIQE: 3.714298, Corrected NIQE: 3.816586
IMG-20240702-WA0045.jpg - Original NIQE: 3.315792, Corrected NIQE: 3.376567
IMG-20240702-WA0048.jpg - Original NIQE: 4.558805, Corrected NIQE: 3.865279
IMG-20240702-WA0052.jpg - Original NIQE: 4.020161, Corrected NIQE: 4.199352
IMG-20240702-WA0055.jpg - Original NIQE: 4.660810, Corrected NIQE: 4.577321
IMG-20240702-WA0059.jpg - Original NIQE: 17.955882, Corrected NIQE: 17.163367
IMG-20240702-WA0062.jpg - Original NIQE: 4.371190, Corrected NIQE: 4.402427
IMG-20240702-WA0064.jpg - Original NIQE: 3.610530, Corrected NIQE: 3.523166
img1.jpg - Original NIQE: 4.889993, Corrected NIQE: 5.309626
img10.jpg - Original NIQE: 2.749435, Corrected NIQE: 2.840842
img13.jpg - Original NIQE: 3.258756, Corrected NIQE: 3.338164
img15.jpg - Original NIQE: 3.396101, Corrected NIQE: 3.493868
img16.jpg - Original NIQE: 3.141336, Corrected NIQE: 2.861115
img24.jpg - Original NIQE: 5.247923, Corrected NIQE: 5.386150
img25.jpg - Original NIQE: 3.937426, Corrected NIQE: 3.625347
img26.jpg - Original NIQE: 3.292083, Corrected NIQE: 3.522594
img28.jpg - Original NIQE: 3.318365, Corrected NIQE: 3.455410
img29.jpg - Original NIQE: 2.959634, Corrected NIQE: 3.154294
img3.jpg - Original NIQE: 3.182477, Corrected NIQE: 3.263272
img32.jpg - Original NIQE: 4.822695, Corrected NIQE: 5.250134
img33.jpg - Original NIQE: 7.346191, Corrected NIQE: 8.489676
img34.jpg - Original NIQE: 5.026021, Corrected NIQE: 4.842230
img35.jpg - Original NIQE: 3.935253, Corrected NIQE: 3.651541
img36.jpg - Original NIQE: 3.207646, Corrected NIQE: 2.831477
img37.jpg - Original NIQE: 3.829129, Corrected NIQE: 3.282875
img38.jpg - Original NIQE: 4.278388, Corrected NIQE: 4.098264
img39.jpg - Original NIQE: 3.990222, Corrected NIQE: 3.693769
img9.jpg - Original NIQE: 3.105263, Corrected NIQE: 3.082071

```

```

total_images = length(niqe_scores_corrected);
fprintf('Number of images with improved NIQE: %d out of %d\n', improvement_count,
total_images);

```

Number of images with improved NIQE: 45 out of 115

```

improvement_percentage = (improvement_count / total_images) * 100;
fprintf('Percentage of images with improved perceptual quality: %.2f%%\n',
improvement_percentage);

```

Percentage of images with improved perceptual quality: 39.13%

```

disp('-----');

```

```
-----  
disp('----- BONUS POINT -----');
```

```
----- BONUS POINT -----
```

```
disp('----- 2nd apporach -----');
```

```
----- 2nd apporach -----
```

```
disp('-----');
```

```
-----
```

```
folderPath = subfolders{2};  
imgFiles = dir(fullfile(folderPath, '*.jpg'));  
  
entropy_values = zeros(length(imgFiles), 1);  
kurtosis_values = zeros(length(imgFiles), 1);  
  
function [entropy_val, kurtosis_val] = compute_metrics(img)  
    img_double = im2double(img);  
    img_flattened = img_double(:);  
  
    entropy_val = entropy(img_flattened);  
    kurtosis_val = kurtosis(img_flattened);  
end  
  
for i = 1:length(imgFiles)  
    imgPath = fullfile(folderPath, imgFiles(i).name);  
    img = imread(imgPath);  
  
    [entropy_values(i), kurtosis_values(i)] = compute_metrics(img);  
end  
  
average_entropy = mean(entropy_values);  
average_kurtosis = mean(kurtosis_values);  
  
fprintf('Desired Average Entropy: %f\n', average_entropy);
```

```
Desired Average Entropy: 7.341499
```

```
fprintf('Desired Average Kurtosis: %f\n', average_kurtosis);
```

```
Desired Average Kurtosis: 2.901754
```

```
entropy_after_values = [];  
kurtosis_after_values = [];  
  
entropy_after_values = [];  
kurtosis_after_values = [];
```

```

output_folder = 'gamma_corrected';
for i = 1:length(subfolders)
    folderPath = subfolders{i};
    imgFiles = dir(fullfile(folderPath, '*.jpg'));
    for j = 1:length(imgFiles)
        imgPath = fullfile(folderPath, imgFiles(j).name);
        img = imread(imgPath);

        [entropy_before, kurtosis_before] = compute_metrics(img);

        % apply gamma correction
        %img_corrected = applyGammaCorrection(im2double(img), 2.2); % test
        correctedImagePath = fullfile(output_folder, subfolders{i},
[imgFiles(j).name(1:end-4), '_correct.jpg']);
        img_corrected = imread(correctedImagePath);

        [entropy_after, kurtosis_after] = compute_metrics(img_corrected);

        entropy_after_values(end+1) = entropy_after;
        kurtosis_after_values(end+1) = kurtosis_after;

        fprintf('File: %s\n', imgFiles(j).name);
        fprintf('Entropy Before: %f, After: %f\n', entropy_before, entropy_after);
        fprintf('Kurtosis Before: %f, After: %f\n', kurtosis_before,
kurtosis_after);
    end
end

```

```

File: IMG-20240702-WA0022.jpg
Entropy Before: 5.852791, After: 5.993950
Kurtosis Before: 2.621677, After: 3.375220
File: IMG-20240702-WA0024.jpg
Entropy Before: 5.376990, After: 5.490250
Kurtosis Before: 3.601264, After: 4.591066
File: IMG-20240702-WA0028.jpg
Entropy Before: 5.274485, After: 5.581996
Kurtosis Before: 1.805720, After: 1.891430
File: IMG-20240702-WA0031.jpg
Entropy Before: 4.763301, After: 5.268005
Kurtosis Before: 10.668817, After: 6.806195
File: IMG-20240702-WA0034.jpg
Entropy Before: 5.253133, After: 5.427970
Kurtosis Before: 3.628491, After: 4.164847
File: IMG-20240702-WA0037.jpg
Entropy Before: 4.874921, After: 5.181920
Kurtosis Before: 2.22826, After: 2.448236
File: IMG-20240702-WA0040.jpg
Entropy Before: 5.012913, After: 5.227319
Kurtosis Before: 4.499397, After: 6.013143
File: IMG-20240702-WA0043.jpg
Entropy Before: 5.868937, After: 5.979437
Kurtosis Before: 2.398634, After: 2.528797
File: IMG-20240702-WA0046.jpg
Entropy Before: 5.917778, After: 6.118015
Kurtosis Before: 7.907488, After: 5.126669
File: IMG-20240702-WA0049.jpg

```

Entropy Before: 5.996696, After: 6.157711
 Kurtosis Before: 2.416991, After: 2.729328
 File: IMG-20240702-WA0051.jpg
 Entropy Before: 5.544875, After: 5.793458
 Kurtosis Before: 13.295061, After: 7.821021
 File: IMG-20240702-WA0054.jpg
 Entropy Before: 5.259738, After: 5.699870
 Kurtosis Before: 6.249124, After: 4.455271
 File: IMG-20240702-WA0057.jpg
 Entropy Before: 6.402786, After: 6.613680
 Kurtosis Before: 2.023790, After: 1.683464
 File: IMG-20240702-WA0058.jpg
 Entropy Before: 6.122913, After: 6.189079
 Kurtosis Before: 14.450236, After: 7.146068
 File: IMG-20240702-WA0063.jpg
 Entropy Before: 4.769043, After: 4.874077
 Kurtosis Before: 6.466660, After: 6.962424
 File: img1.jpg
 Entropy Before: 6.261443, After: 6.477824
 Kurtosis Before: 1.796222, After: 1.755170
 File: img11.jpg
 Entropy Before: 6.344929, After: 6.600959
 Kurtosis Before: 6.610034, After: 3.849122
 File: img12.jpg
 Entropy Before: 5.823728, After: 5.982450
 Kurtosis Before: 9.443453, After: 3.267547
 File: img13.jpg
 Entropy Before: 6.562053, After: 6.672362
 Kurtosis Before: 2.779451, After: 2.055755
 File: img14.jpg
 Entropy Before: 5.461994, After: 5.871466
 Kurtosis Before: 3.008570, After: 2.239606
 File: img15.jpg
 Entropy Before: 5.621420, After: 5.871202
 Kurtosis Before: 20.377873, After: 4.912260
 File: img19.jpg
 Entropy Before: 5.604880, After: 5.926515
 Kurtosis Before: 2.400788, After: 1.900112
 File: img2.jpg
 Entropy Before: 5.557801, After: 5.768281
 Kurtosis Before: 1.549492, After: 1.499915
 File: img20.jpg
 Entropy Before: 5.152422, After: 5.330660
 Kurtosis Before: 4.798603, After: 2.926399
 File: img25.jpg
 Entropy Before: 5.641919, After: 6.017405
 Kurtosis Before: 5.963815, After: 4.464755
 File: img26.jpg
 Entropy Before: 6.567799, After: 6.613218
 Kurtosis Before: 10.450691, After: 4.888727
 File: img27.jpg
 Entropy Before: 5.382264, After: 5.540807
 Kurtosis Before: 3.029250, After: 3.563513
 File: img28.jpg
 Entropy Before: 5.182328, After: 5.513094
 Kurtosis Before: 5.590864, After: 3.703015
 File: img29.jpg
 Entropy Before: 6.385328, After: 6.351211
 Kurtosis Before: 8.555310, After: 5.234314
 File: img30.jpg
 Entropy Before: 5.842811, After: 6.091731
 Kurtosis Before: 4.107055, After: 3.210364
 File: img31.jpg
 Entropy Before: 6.120935, After: 6.329188

Kurtosis Before: 2.329751, After: 2.224626
 File: img32.jpg
 Entropy Before: 7.176206, After: 7.188640
 Kurtosis Before: 2.034253, After: 1.804649
 File: img33.jpg
 Entropy Before: 5.949585, After: 6.043028
 Kurtosis Before: 1.347242, After: 1.249411
 File: img34.jpg
 Entropy Before: 4.057834, After: 4.489283
 Kurtosis Before: 16.333946, After: 12.476384
 File: img35.jpg
 Entropy Before: 6.820647, After: 6.891656
 Kurtosis Before: 1.633271, After: 1.656331
 File: img37.jpg
 Entropy Before: 6.321628, After: 6.434378
 Kurtosis Before: 1.189132, After: 1.209155
 File: img38.jpg
 Entropy Before: 6.332264, After: 6.445936
 Kurtosis Before: 1.461115, After: 1.455146
 File: img39.jpg
 Entropy Before: 6.530569, After: 6.724030
 Kurtosis Before: 6.036804, After: 2.984971
 File: img8.jpg
 Entropy Before: 6.137960, After: 6.194554
 Kurtosis Before: 11.047255, After: 7.327304
 File: IMG-20240702-WA0023.jpg
 Entropy Before: 7.379780, After: 7.467988
 Kurtosis Before: 3.227124, After: 2.934454
 File: IMG-20240702-WA0026.jpg
 Entropy Before: 7.032249, After: 7.128946
 Kurtosis Before: 4.355580, After: 3.870211
 File: IMG-20240702-WA0029.jpg
 Entropy Before: 7.286447, After: 7.271493
 Kurtosis Before: 2.062579, After: 2.088563
 File: IMG-20240702-WA0032.jpg
 Entropy Before: 7.016490, After: 7.035809
 Kurtosis Before: 3.765748, After: 3.495361
 File: IMG-20240702-WA0035.jpg
 Entropy Before: 6.994351, After: 7.029026
 Kurtosis Before: 4.007345, After: 3.846024
 File: IMG-20240702-WA0038.jpg
 Entropy Before: 6.813101, After: 6.830390
 Kurtosis Before: 2.718570, After: 2.709822
 File: IMG-20240702-WA0041.jpg
 Entropy Before: 6.873232, After: 6.935915
 Kurtosis Before: 6.056899, After: 5.373508
 File: IMG-20240702-WA0044.jpg
 Entropy Before: 7.125989, After: 7.099738
 Kurtosis Before: 2.188307, After: 2.281163
 File: IMG-20240702-WA0047.jpg
 Entropy Before: 7.413877, After: 7.366222
 Kurtosis Before: 3.044025, After: 2.889486
 File: IMG-20240702-WA0050.jpg
 Entropy Before: 7.549739, After: 7.619864
 Kurtosis Before: 2.348080, After: 2.218618
 File: IMG-20240702-WA0053.jpg
 Entropy Before: 7.309356, After: 7.305213
 Kurtosis Before: 3.641825, After: 3.689264
 File: IMG-20240702-WA0056.jpg
 Entropy Before: 7.795790, After: 7.795729
 Kurtosis Before: 1.910903, After: 1.910789
 File: IMG-20240702-WA0060.jpg
 Entropy Before: 7.181877, After: 7.156604
 Kurtosis Before: 3.779969, After: 3.828454

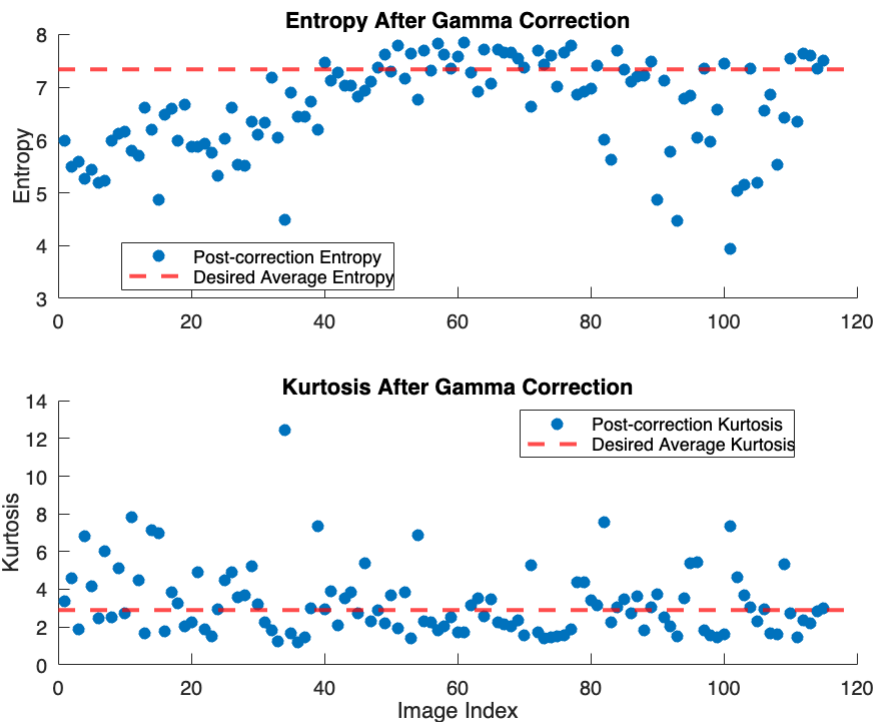
File: IMG-20240702-WA0061.jpg
Entropy Before: 7.655906, After: 7.646593
Kurtosis Before: 1.398567, After: 1.417687
File: IMG-20240702-WA0065.jpg
Entropy Before: 6.681183, After: 6.763327
Kurtosis Before: 7.491910, After: 6.873147
File: img10.jpg
Entropy Before: 7.694504, After: 7.696649
Kurtosis Before: 2.312457, After: 2.310516
File: img12.jpg
Entropy Before: 7.241292, After: 7.307457
Kurtosis Before: 2.431026, After: 2.244382
File: img13.jpg
Entropy Before: 7.822392, After: 7.831179
Kurtosis Before: 1.841072, After: 1.841438
File: img14.jpg
Entropy Before: 7.619292, After: 7.618956
Kurtosis Before: 2.058350, After: 2.057390
File: img15.jpg
Entropy Before: 7.352590, After: 7.352613
Kurtosis Before: 2.537156, After: 2.536913
File: img18.jpg
Entropy Before: 7.521841, After: 7.573052
Kurtosis Before: 1.836094, After: 1.712975
File: img19.jpg
Entropy Before: 7.884117, After: 7.849232
Kurtosis Before: 1.746999, After: 1.732019
File: img21.jpg
Entropy Before: 7.329321, After: 7.283210
Kurtosis Before: 3.245348, After: 3.149586
File: img24.jpg
Entropy Before: 6.917942, After: 6.918481
Kurtosis Before: 3.512095, After: 3.512542
File: img25.jpg
Entropy Before: 7.747956, After: 7.705599
Kurtosis Before: 2.721685, After: 2.585344
File: img27.jpg
Entropy Before: 7.016036, After: 7.060299
Kurtosis Before: 3.984167, After: 3.446638
File: img29.jpg
Entropy Before: 7.719325, After: 7.722338
Kurtosis Before: 2.244502, After: 2.243832
File: img30.jpg
Entropy Before: 7.675243, After: 7.649763
Kurtosis Before: 2.141504, After: 2.134456
File: img31.jpg
Entropy Before: 7.652320, After: 7.652173
Kurtosis Before: 2.071630, After: 2.045032
File: img32.jpg
Entropy Before: 7.572325, After: 7.552300
Kurtosis Before: 2.743317, After: 2.368640
File: img33.jpg
Entropy Before: 7.409284, After: 7.364097
Kurtosis Before: 1.598484, After: 1.547056
File: img34.jpg
Entropy Before: 6.665231, After: 6.624742
Kurtosis Before: 5.583943, After: 5.257143
File: img35.jpg
Entropy Before: 7.688103, After: 7.699824
Kurtosis Before: 1.766044, After: 1.705478
File: img36.jpg
Entropy Before: 7.423673, After: 7.422093
Kurtosis Before: 1.407104, After: 1.380184
File: img37.jpg

Entropy Before: 7.628640, After: 7.608669
 Kurtosis Before: 1.531837, After: 1.443935
 File: img38.jpg
 Entropy Before: 6.968815, After: 7.004277
 Kurtosis Before: 1.536791, After: 1.493204
 File: img39.jpg
 Entropy Before: 7.658482, After: 7.666796
 Kurtosis Before: 1.641161, After: 1.580328
 File: img4.jpg
 Entropy Before: 7.778323, After: 7.781281
 Kurtosis Before: 1.869212, After: 1.880458
 File: img5.jpg
 Entropy Before: 6.909359, After: 6.851768
 Kurtosis Before: 4.606276, After: 4.353655
 File: img6.jpg
 Entropy Before: 6.922219, After: 6.913119
 Kurtosis Before: 4.402954, After: 4.347654
 File: img7.jpg
 Entropy Before: 6.965542, After: 6.966136
 Kurtosis Before: 3.409538, After: 3.399302
 File: img8.jpg
 Entropy Before: 7.449433, After: 7.418026
 Kurtosis Before: 3.095499, After: 3.145044
 File: IMG-20240702-WA0025.jpg
 Entropy Before: 5.362200, After: 6.009268
 Kurtosis Before: 11.490008, After: 7.561884
 File: IMG-20240702-WA0027.jpg
 Entropy Before: 5.477689, After: 5.622062
 Kurtosis Before: 3.191516, After: 2.233324
 File: IMG-20240702-WA0030.jpg
 Entropy Before: 7.669052, After: 7.688023
 Kurtosis Before: 2.996009, After: 3.032724
 File: IMG-20240702-WA0033.jpg
 Entropy Before: 6.806307, After: 7.337793
 Kurtosis Before: 7.419890, After: 3.449207
 File: IMG-20240702-WA0036.jpg
 Entropy Before: 6.707891, After: 7.100683
 Kurtosis Before: 3.245324, After: 2.739654
 File: IMG-20240702-WA0039.jpg
 Entropy Before: 6.561325, After: 7.207241
 Kurtosis Before: 6.391606, After: 3.612551
 File: IMG-20240702-WA0042.jpg
 Entropy Before: 6.760973, After: 7.213502
 Kurtosis Before: 2.897808, After: 1.831878
 File: IMG-20240702-WA0045.jpg
 Entropy Before: 7.448186, After: 7.487940
 Kurtosis Before: 2.984482, After: 3.065789
 File: IMG-20240702-WA0048.jpg
 Entropy Before: 4.300470, After: 4.876541
 Kurtosis Before: 4.805696, After: 3.738848
 File: IMG-20240702-WA0052.jpg
 Entropy Before: 6.280692, After: 7.122232
 Kurtosis Before: 7.222053, After: 2.495105
 File: IMG-20240702-WA0055.jpg
 Entropy Before: 5.157346, After: 5.774853
 Kurtosis Before: 2.369418, After: 2.009678
 File: IMG-20240702-WA0059.jpg
 Entropy Before: 4.140669, After: 4.466448
 Kurtosis Before: 1.689197, After: 1.496316
 File: IMG-20240702-WA0062.jpg
 Entropy Before: 6.184884, After: 6.786718
 Kurtosis Before: 5.258829, After: 3.498820
 File: IMG-20240702-WA0064.jpg
 Entropy Before: 6.761156, After: 6.850947

Kurtosis Before: 8.463568, After: 5.371459
 File: img1.jpg
 Entropy Before: 5.529793, After: 6.035903
 Kurtosis Before: 7.746104, After: 5.428856
 File: img10.jpg
 Entropy Before: 7.082011, After: 7.358574
 Kurtosis Before: 2.506676, After: 1.802527
 File: img13.jpg
 Entropy Before: 5.661195, After: 5.972191
 Kurtosis Before: 2.328716, After: 1.577517
 File: img15.jpg
 Entropy Before: 6.354024, After: 6.579448
 Kurtosis Before: 2.215675, After: 1.474276
 File: img16.jpg
 Entropy Before: 7.109905, After: 7.443516
 Kurtosis Before: 1.816291, After: 1.622918
 File: img24.jpg
 Entropy Before: 3.203539, After: 3.932380
 Kurtosis Before: 8.874162, After: 7.356168
 File: img25.jpg
 Entropy Before: 4.199050, After: 5.030343
 Kurtosis Before: 5.997876, After: 4.638101
 File: img26.jpg
 Entropy Before: 4.596796, After: 5.150683
 Kurtosis Before: 4.901364, After: 3.699422
 File: img28.jpg
 Entropy Before: 6.896679, After: 7.353151
 Kurtosis Before: 4.778826, After: 3.031709
 File: img29.jpg
 Entropy Before: 4.899033, After: 5.184862
 Kurtosis Before: 3.143767, After: 2.283323
 File: img3.jpg
 Entropy Before: 6.068055, After: 6.555881
 Kurtosis Before: 4.146524, After: 2.948114
 File: img32.jpg
 Entropy Before: 6.497520, After: 6.857223
 Kurtosis Before: 3.015591, After: 1.654807
 File: img33.jpg
 Entropy Before: 5.509605, After: 5.530238
 Kurtosis Before: 2.700859, After: 1.598981
 File: img34.jpg
 Entropy Before: 6.465218, After: 6.425248
 Kurtosis Before: 5.801966, After: 5.316840
 File: img35.jpg
 Entropy Before: 7.296126, After: 7.535968
 Kurtosis Before: 3.036497, After: 2.700128
 File: img36.jpg
 Entropy Before: 6.107121, After: 6.355487
 Kurtosis Before: 2.072370, After: 1.430801
 File: img37.jpg
 Entropy Before: 7.182883, After: 7.641842
 Kurtosis Before: 3.493271, After: 2.356650
 File: img38.jpg
 Entropy Before: 7.112011, After: 7.594447
 Kurtosis Before: 2.629882, After: 2.173193
 File: img39.jpg
 Entropy Before: 7.108924, After: 7.359004
 Kurtosis Before: 2.750256, After: 2.834465
 File: img9.jpg
 Entropy Before: 7.196257, After: 7.496897
 Kurtosis Before: 2.776226, After: 2.997075

Plot entropy and kurtosis

```
figure;  
subplot(2,1,1);  
scatter(1:length(entropy_after_values), entropy_after_values, 'filled');  
hold on;  
yline(average_entropy, 'r--', 'LineWidth', 2);  
ylabel('Entropy');  
title('Entropy After Gamma Correction');  
legend('Post-correction Entropy', 'Desired Average Entropy', 'Location', 'best');  
  
subplot(2,1,2);  
scatter(1:length(kurtosis_after_values), kurtosis_after_values, 'filled');  
hold on;  
yline(average_kurtosis, 'r--', 'LineWidth', 2);  
ylabel('Kurtosis');  
title('Kurtosis After Gamma Correction');  
legend('Post-correction Kurtosis', 'Desired Average Kurtosis', 'Location', 'best');  
  
xlabel('Image Index');
```



Check results

```
function plotImageAndHistogram(output_folder, subfolder, imageName)  
    originalImagePath = fullfile(subfolder, imageName);  
    correctedImagePath = fullfile(output_folder, subfolder, [imageName(1:end-4),  
    '_correct.jpg']);
```

```

originalImg = imread(originalImagePath);
correctedImg = imread(correctedImagePath);

originalYCbCr = rgb2ycbcr(originalImg);
correctedYCbCr = rgb2ycbcr(correctedImg);

originalY = originalYCbCr(:,:,1);
correctedY = correctedYCbCr(:,:,1);

originalYHist = imhist(originalY);
correctedYHist = imhist(correctedY);

figure;
subplot(2, 2, 1);
imshow(originalImg);
title('Original Image');

subplot(2, 2, 2);
bar(originalYHist, 'r');
title('Y Histogram of Original Image');
xlim([0 256]);

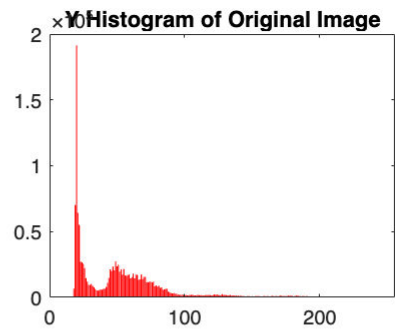
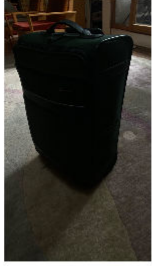
subplot(2, 2, 3);
imshow(correctedImg);
title('Gamma Corrected Image');

subplot(2, 2, 4);
bar(correctedYHist, 'b');
title('Y Histogram of Corrected Image');
xlim([0 256]);
end

plotImageAndHistogram(output_folder, 'underexposed', 'img39.jpg');

```

Original Image



Gamma Corrected Image

